

08.01.2026.

$$* \quad v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}, \quad v_4 = \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}$$

$$v_1 \cdot v_1 = 4, \quad v_2 \cdot v_2 = 4, \quad v_3 \cdot v_3 = 4, \quad v_4 \cdot v_4 = 4$$

All 4 vectors are orthogonal but not orthonormal.

$$b = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$$

$$b = \underbrace{(b \cdot \vec{\varphi}_1)}_1 \vec{\varphi}_1 + \underbrace{(b \cdot \vec{\varphi}_2)}_1 \vec{\varphi}_2 + \dots + \underbrace{(b \cdot \vec{\varphi}_n)}_1 \vec{\varphi}_n$$

If $\varphi_1, \varphi_2, \dots, \varphi_n$ are ON

$$= \frac{b \cdot \varphi_1}{\varphi_1 \cdot \varphi_1} \vec{\varphi}_1 + \frac{b \cdot \varphi_2}{\varphi_2 \cdot \varphi_2} \vec{\varphi}_2 + \dots + \frac{b \cdot \varphi_n}{\varphi_n \cdot \varphi_n} \vec{\varphi}_n$$

$$\begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} = \frac{\begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}}{4} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} + \frac{\begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}}{4} \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix} + \dots$$

$$* A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Find N_A & $\text{Col.Sp}(A)$. What do you observe? Can you generalize it for a square matrix?

Soln N_A : $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} x_1 = k \\ x_2 = 0 \end{matrix} \therefore N_A = \left\{ k \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$

to $Ax = b$

$\text{Col.Sp}(A)$: Only l.i. col of A : $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow \text{Col.Sp}(A) = \left\{ k \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$

For the given A , N_A & $\text{Col.Sp}(A)$ are the same.

Given A , what is the $\text{col.Sp}(A) = ?$

$$\begin{aligned} &= \vec{b} = x_1 \vec{A}_1 + x_2 \vec{A}_2 + \dots + x_n \vec{A}_n \\ &= \text{Span of } \text{col.Sp}(A) \end{aligned}$$

$$\left\{ \begin{array}{l} \vec{b} = A\vec{x} \\ \vec{b} \in \mathbb{R}^n \end{array} \right\} \rightarrow \text{col. sp}(A).$$

for a sq. matrix A .

$A^{n \times n}$ col. space
dim subs
of $\mathbb{R}^n$

$$N_A: A\vec{y} = \vec{0} \quad y \in N_A \text{ if } A\vec{y} = \vec{0}$$

Because N_A & $\text{ColSp}(A)$ are the same

$$\Rightarrow A(\underline{A\vec{x}}) = \underline{\vec{0}}$$

$$= A^2 \vec{x} = \vec{0}$$

x is any vector in $\mathbb{R}^n$
 $A^{n \times n}$.

$$A^2 = 0 \text{ Nilpotent}$$

matrices. have $N_A = \text{Col. Sp}(A)$.

$$\Rightarrow \text{rank}(A) = \text{rank}(A).$$

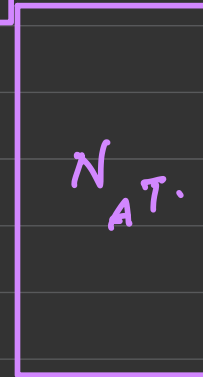
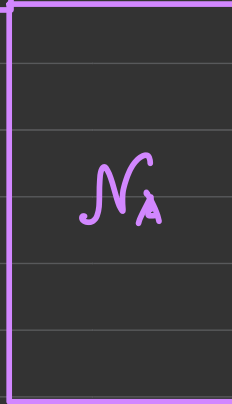
Nilpotent matrices of even dimension with $\text{rank}(A) = \frac{n}{2}$

$$\text{will have } N_A = \text{Col. Sp}(A) \Rightarrow \text{rank}(A) = \frac{n}{2}.$$

$\mathbb{R}^n$

$A^{n \times n}$

$\mathbb{R}^n$



Orthonormalizⁿ using Gram-Schmidt Procedure.

Let u, v, w be a l.i set of vectors & let

e, f, g be orthogonal vectors generated from u, v, w .

Q1: How do we generate e, f, g from u, v, w ?

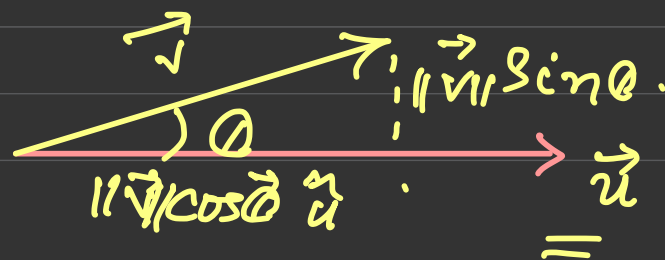
Q2. How do we know that e, f, g span the same vector space as u, v, w ?

Q1: How do we generate e, f, g from u, v, w ?

To answer this, let us just look at two

vectors u & v .

$\{u, v\}$



Choose $\vec{u}$ to be the reference vector.

$$\left\{ \begin{array}{l} \vec{e} = \vec{u} \\ \vec{f} = \vec{v} - \text{Proj}_{\vec{u}} \vec{v} \\ \quad = \vec{v} - \frac{\vec{v} \cdot \vec{u}}{\vec{u} \cdot \vec{u}} \vec{u} \\ \vec{g} = \vec{w} - \text{Proj}_{\vec{f}} \vec{w} - \text{Proj}_{\vec{e}} \vec{w} \\ \quad = \vec{w} - \frac{\vec{w} \cdot \vec{f}}{\vec{f} \cdot \vec{f}} \vec{f} - \frac{\vec{w} \cdot \vec{e}}{\vec{e} \cdot \vec{e}} \vec{e} \end{array} \right.$$

$$\begin{aligned} \vec{e} &= \vec{u} \\ \vec{e} \cdot \vec{f} &= \vec{u} \cdot \left(\vec{v} - \frac{\vec{v} \cdot \vec{u}}{\vec{u} \cdot \vec{u}} \vec{u} \right) \\ &= \vec{u} \cdot \vec{v} - \frac{\vec{v} \cdot \vec{u}}{\vec{u} \cdot \vec{u}} (\vec{u} \cdot \vec{u}) \\ &= 0 \end{aligned}$$

$$\text{Ex: } \vec{u} = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix} \quad \vec{v} = \begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix} \quad \vec{w} = \begin{pmatrix} 5 \\ 4 \\ 3 \end{pmatrix}.$$

$\vec{e}, \vec{f}, \vec{g}$

$$\vec{e} = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}$$

$$\vec{f} = \begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix} - \frac{(4 \ 3 \ 5)}{50} \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}$$

$$= \begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix} - \frac{49}{50} \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix} = \begin{pmatrix} 53/50 \\ -46/50 \\ 5/50 \end{pmatrix}$$

$$\vec{g} = ?$$

Q2. Do $\vec{e}, \vec{f}, \vec{g}$ span the same space as $\vec{u}, \vec{v}, \vec{w}$?

$$\vec{e} = \vec{u}$$

$$\Rightarrow \vec{u} = \vec{e}$$

$$\vec{f} = \vec{v} - \frac{\vec{v} \cdot \vec{u}}{\vec{u} \cdot \vec{u}} \vec{e}$$

$$\vec{v} = \vec{f} + \frac{\vec{v} \cdot \vec{u}}{\vec{u} \cdot \vec{u}} \vec{e}$$

$$\vec{g} = \vec{w} - \frac{\vec{w} \cdot \vec{u}}{\vec{u} \cdot \vec{u}} \vec{e} - \frac{\vec{w} \cdot \vec{f}}{\vec{f} \cdot \vec{f}} \vec{f}$$

$$\vec{w} = \vec{g} + \frac{\vec{w} \cdot \vec{u}}{\vec{u} \cdot \vec{u}} \vec{e} + \frac{\vec{w} \cdot \vec{f}}{\vec{f} \cdot \vec{f}} \vec{f}$$

$\vec{e}, \vec{f}, \vec{g}$ are orthogonal & hence l.i., $\vec{u}, \vec{v}, \vec{w}$ are l.i.

$$\vec{x} \in \text{Span of } \underline{\vec{u}, \vec{v}, \vec{w}}$$

$$= \vec{x} = \alpha \vec{u} + \beta \vec{v} + \gamma \vec{w} \quad \text{for } \alpha, \beta, \gamma \text{ scalars.}$$

$$= \alpha(\vec{e}) + \beta\left(\vec{f} + \frac{\vec{v} \cdot \vec{u}}{\vec{u} \cdot \vec{u}} \vec{e}\right) + \gamma\left(\vec{g} + \frac{\vec{w} \cdot \vec{u}}{\vec{u} \cdot \vec{u}} \vec{e} + \frac{\vec{w} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \vec{f}\right)$$

$$= \underbrace{\left(\alpha + \beta\left(\frac{\vec{v} \cdot \vec{u}}{\vec{u} \cdot \vec{u}}\right) + \gamma\left(\frac{\vec{w} \cdot \vec{u}}{\vec{u} \cdot \vec{u}}\right)\right)}_{k_1} \vec{e} + \underbrace{\left(\beta + \frac{\vec{w} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \gamma\right)}_{k_2} \vec{f}$$

$$\vec{x} = \underset{=}{k_1} \underset{=}{\vec{e}} + \underset{=}{k_2} \underset{=}{\vec{f}} + \underset{=}{k_3} \underset{=}{\vec{g}} + \cancel{\gamma \vec{g}}$$

Ex: 2. $\vec{u} = \begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix}$ $\vec{v} = \begin{pmatrix} 0 \\ 3 \\ 4 \end{pmatrix}$ $\vec{w} = \begin{pmatrix} 3 \\ 7 \\ 4 \end{pmatrix}$.

GS. $\Rightarrow \vec{e} = \begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix}$ $\vec{f} = \begin{pmatrix} -36/25 \\ 27/25 \\ 4 \end{pmatrix}$ $\vec{g} = \begin{pmatrix} 10/10 \\ 10/10 \\ 0 \end{pmatrix}$

$$\vec{q} = \begin{pmatrix} 3 \\ 7 \\ 4 \end{pmatrix} - \underbrace{(3 \ 7 \ 4) \begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix}}_{(3 \ 4 \ 0) \begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix}} - \underbrace{(3 \ 7 \ 4) \begin{pmatrix} -36/25 \\ 27/25 \\ 4 \end{pmatrix}}_{(-36/25 \ \frac{27}{25} \ 4) \begin{pmatrix} -36/25 \\ 27/25 \\ 4 \end{pmatrix}} \begin{pmatrix} -36/25 \\ 27/25 \\ 4 \end{pmatrix}$$

$-108 + 189 + 400$

$$= \begin{pmatrix} 3 \\ 7 \\ 4 \end{pmatrix} - \frac{37}{25} \begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix} - \frac{481/25}{\frac{1296}{625} + \frac{729}{625} + \frac{10000}{625}} \begin{pmatrix} -36/25 \\ 27/25 \\ 100/25 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \\ 7 \\ 4 \end{pmatrix} - \frac{37}{25} \begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix} + \left(\frac{481 \times 36}{-481 \times 27} + \frac{481 \times 100}{12025} \right)$$

$$= \begin{pmatrix} 3 \\ 7 \\ 4 \end{pmatrix} - \frac{37}{25} \begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix} - \begin{pmatrix} -36/25 \\ 27/25 \\ 100/25 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

GSO is also a test for linear independence.

If GSO results in a zero vector, it means, the set is a linearly dep set of vectors.